Designing a Photo-Sharing App Database

Lecture Notes · Game DB Junior

In this lesson

- Three tips for deciding **what tables** to make
- How to find **resources** and **relationships** by looking at app screens
- The first design: **users, photos, comments, likes**

1. The four tables

In the last lesson we said our photo sharing app database should have **four tables**: a list of users, a list of photos, a list of comments and a list of likes.

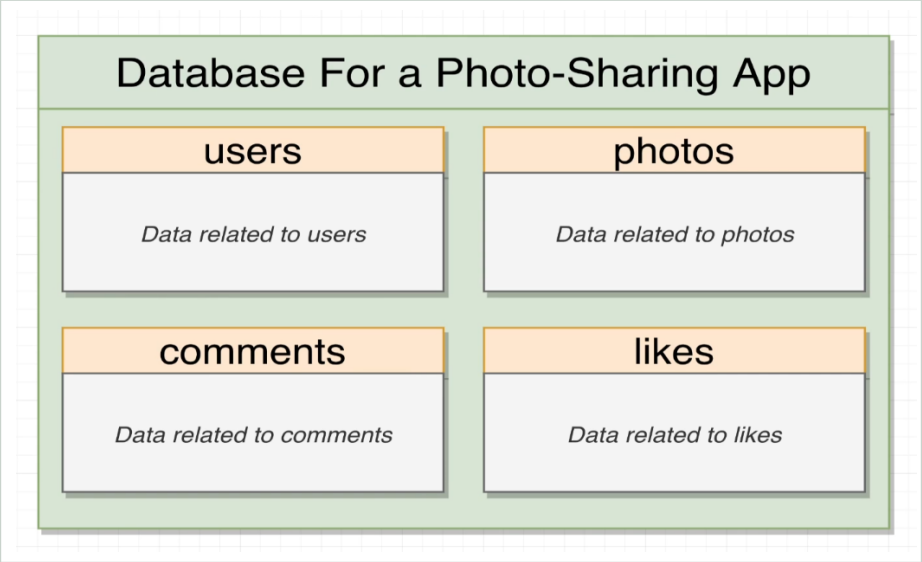


Figure 1 — Database for a photo-sharing app

What are likes? In some apps you can upvote a post or like a photo. The **likes** table is for building this liking feature in our photo sharing app.

But how did we come up with these four tables? Here are some tips for making your own database designs.

2. What tables should we make?

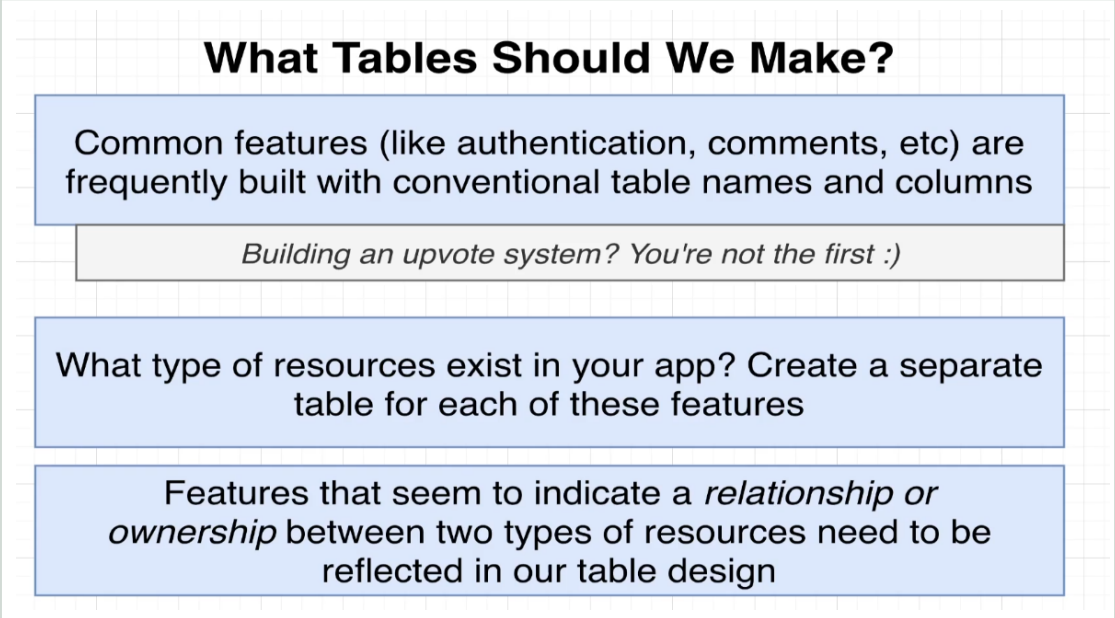


Figure 2 — What tables should we make?

Tip 1: Look for common features

- Many apps have features that are **common** in many other web apps: user **authentication**, a **liking** system, a **commenting** or discussion system.
- These common features are often built with **usual (conventional) table names and columns**.
- There are lots of resources online about how to structure your database for these features.
- Example: you want users to like images or posts. Your first resource is a simple **Google search**, like "SQL schema like system" or "upvote system".
- Building an upvote system? You're not the first. If another app probably has the same feature, Google is your number one resource.

Tip 2: Find the resources in your app

- Sometimes your app has a feature that **no one has built before**.
- Then look at **mockups** (early designs of the user interface) and think about the purpose of your app.
- Try to find the different **types of resources** in your app.
- As a first design, create a **separate table** for each resource.

Tip 3: Find relationships and ownership

- For each resource, look for a **relationship** or **ownership** between it and other resources.
- These relationships need to be shown in our table design.

3. Example: the Instagram app

Words alone are not very useful here; we need to see this in action. So we look at two screenshots from the Instagram mobile app, which is also a photo sharing app.

Screenshot 1: a user profile page

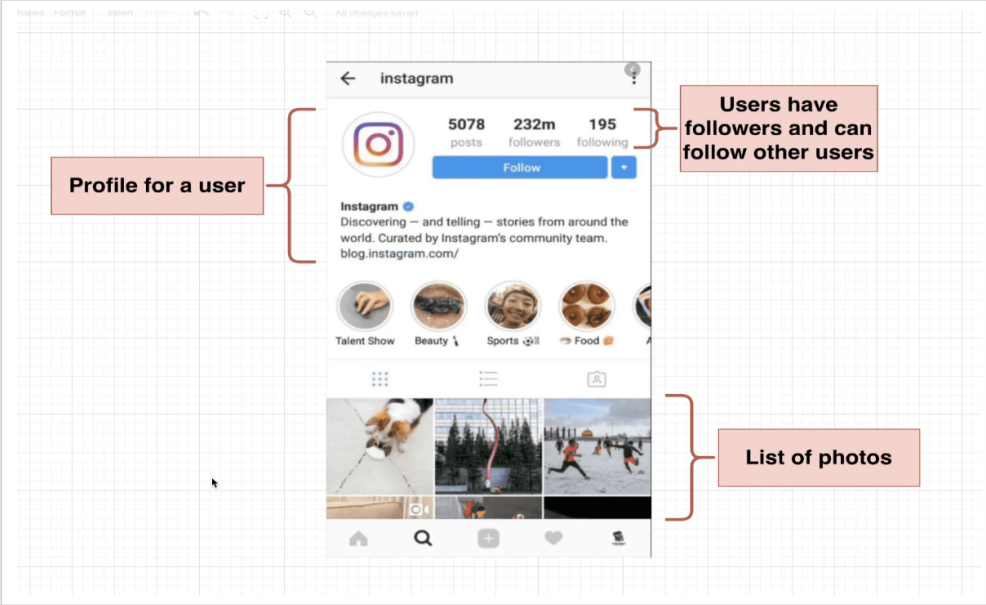


Figure 3 — A user profile page

What we see	What it tells us
A profile for a user	There are **users**. A user has a **user name** (here: Instagram) and a short **biography** text
Followers and following	Users can follow other users and be followed. There is a **relationship between users**
A list of photos	There is a resource (maybe a table) of **photos**. These photos were posted by this user, so a **user owns photos**

Screenshot 2: a single post

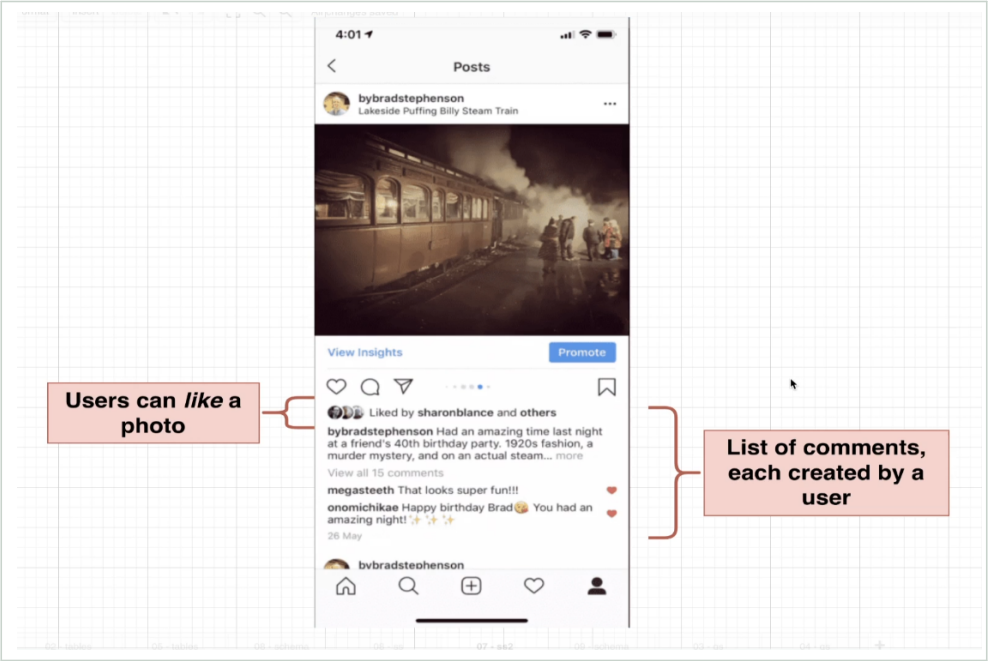


Figure 4 — A single photo (a post)

What we see	What it tells us
A list of comments	There is a **comments** resource stored somewhere in a database
Each comment is created by a user	There is a relationship between **comments and users**
The comments are for this post	There is a relationship between **comments and photos**
Users can like a photo	There is a **likes** resource. There is a relationship between **users and likes**, and between **likes and photos**

4. The first design

From the two screenshots we found four resources and the relationships between them:

- A user creates and owns **photos**.
- A user creates and owns **comments**.
- A user creates and owns **likes**.
- **Photos and likes** are related, because you can like a photo.
- **Photos and comments** are related, because you can add a comment to a photo.

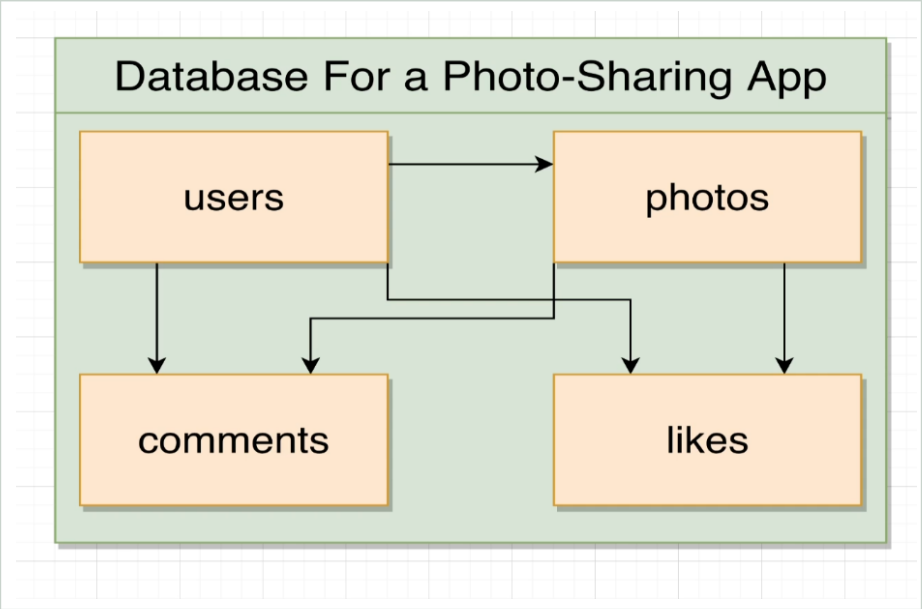


Figure 5 — Relationships between the four tables

Do not worry about what the arrow directions mean yet. The lines show that records in these tables are connected: either a plain **relationship** or some kind of **ownership**.

5. Summary

Idea	Remember
Common features	Search online. Others have built them before
Resources	Find the types of resources in your app. One table for each
Relationships	Find relationships or ownership between resources and show them in the design
Photo app design	users, photos, comments, likes. Users own photos, comments and likes. Photos are related to comments and likes

Next lesson: we take this design one step further.

Sources
Lecture transcript and lecture images (Figures 1–5)